Class Meeting Handout #1

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This document was produced using the LaTeX typesetting language with the Tufte-handout document class.

This document contains information for the discussion and activities of our first Chemistry 4410 class meeting.

Introducing Physical Organic Chemistry

If you combine physical and organic chemistry, do you get physical organic chemistry? It's more than that. It is much greater than the sum of its parts.

Discussion Exercise – 20 minutes

I will welcome you to the course and explain how it's organized. Then we will discuss what physical organic chemistry means to you and how it will be helpful to you. You will have read the preface (pages *xxiii* to *xxiv*) to the textbook[1,2] and Joseph Mullin's opinion piece about the core of organic chemistry[3] as instructed on the Moodle page. Let us now discuss the following question…

[1] "Modern Physical Organic Chemistry" by Ansyln and Dougherty, **2005**, *University Science Books*, Sausalito, California. **You should get access to a copy of the textbook as soon as possible.** There is a copy in the lounge and a copy on reserve at the library. It can also be ordered via the UPEI bookstore.

[2] The front matter for the textbook, including the preface, is available for free at https://uscibooks.aip.org/wp-content/uploads/adfm_new.pdf.

[3] "Six Pillars of Organic Chemistry." Joseph J. Mullins, *J. Chem. Ed.*, **2008**, *85*, 83-97. https://doi.org/10.1021/ed085p83

1. What topics within your previous chemical training do you expect to increase your mastery of in a course in physical organic chemistry?

Review: Conformation in Alkanes

We will consider the transition between the *gauche* and *anti* conformers of butane. Figure 2.6 in the textbook will provide all the information we need for this exercise.[4]

[4] See page 93 of the textbook

Activity: Rotating a Bond – 20 minutes

2. Define the terms *staggered*, *gauche*, *anti* and *eclipsed*. How do the more precise terminologies of *syn/anti* and *clinal/periplanar* compare?[5]
3. Draw the structures of the *gauche* (*synclinal*) and the *anti* (*antiperiplanar*) conformers using Newman projection.
4. Draw a potential energy diagram for the transition from the *gauche* conformer to the *anti* conformer.
5. Add a reaction coordinate to your diagram.
6. Is the change in conformation a chemical reaction? Debate!
7. Is the *eclipsed* conformer a transition state? Debate!
8. Using the information in Figure 2.6, estimate the rate of rotation around the central bond of butane. Comment on the value and what you know about the NMR spectrum of butane.

[5] Want some controversy? Read "Periplanar or Coplanar?" S. Kane, W.H. Hersh, *J. Chem. Ed.*, **2000**, 77, 1366. https://doi.org/10.1021/ed077p1366

The Eyring equation

$$k = \kappa \frac{k_B}{h} T e^{-\frac{\Delta G^{\ddagger}}{RT}}$$

$\kappa = 1$ by definition

$k_B = 1.381 \times 10^{-23}$ J K−1

$h = 6.626 \times 10^{-34}$ J s

$R = 8.314$ J mol^{-1} K^{-1}

Activity: Steric Effects – 10 minutes

Table 2.14 of the textbook[6] collects known values for the difference in free energy, ΔG°, for a substituent moving from the equatorial to the axial position in a cyclohexane ring.

[6] See page 104 of the textbook

9. Draw the two chair conformers of methylcyclohexane and *t*-butylcyclohexane.
10. Use the data in Table 2.14 and calculate the equilibrium constant for the two conformers in each case. Take note of the results. We will be revisiting this idea in the next class meeting.
11. Compare the A-values reported in the table. Do they describe the size of the group? If so, why does ethyl have the same steric effect as methyl? Debate!

$K_{eq} = e^{-\frac{\Delta G^\circ}{RT}}$

Resources

You will get a good review of conformational analysis and steric effects by reading chapter 2.3 of the textbook.[1] Also, chapter 3.6 & 3.7 of McMurry's Organic Chemistry[7] provide a brief overview of the subject at the 2nd-year level. More resources are available on the Moodle site for this class meeting and for lesson #1.[8]

[7] "Organic Chemistry, 10th ed." is available for free from openstax.org at https://openstax.org/details/books/organic-chemistry

[8] You will find resources applicable across the whole lesson on the page for each lesson. Resources relevant to a class meeting will be found on the page for that meeting within each lesson.

Learning Goals

There is more to a class meeting than what we do in 50 minutes of class time. After participating in the above exercises and completing all additional requirements described on the Moodle site, you will have achieved the following...

1. Understand the goals of the field of physical organic chemistry.
2. Navigate the web book to obtain information about lessons and meetings.
3. Reviewed the concepts of alkane structure and conformational analysis.
4. Understand that structure affects energy.

Looking Ahead

No exercise in the course is without value, no matter how trivial.[9] Today, we played with the idea of a reaction coordinate and the true nature of a transition state. We also saw how the equilibrium constant for ring flipping of substituted cyclohexane could be used to score the steric effect of a substituent. We will revisit both of these topics again and again throughout this course.

[9] See "The Karate Kid" (1984) for an example of seemingly pointless exercises that led to ultimate mastery.

Data

The class exercises need data from the textbook. I will present excerpts from the textbook in the next few pages. There is a copy of the textbook in the library on reserve. Use it, read it and share it. Soon I won't be providing data and expecting students to be using the textbook.

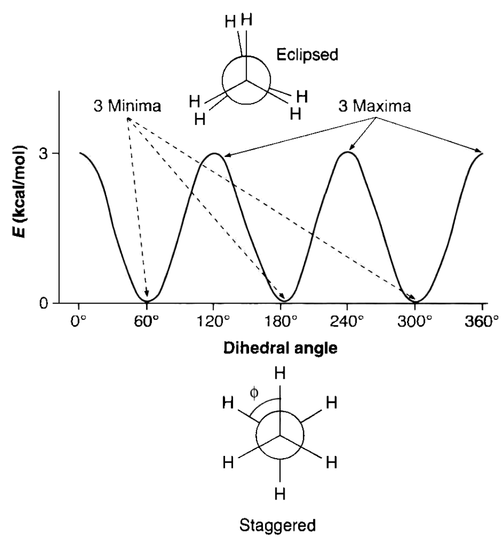

Figure 1: First part of Figure 2.6 in "Modern Physical Organic Chemistry" by Anslyn & Dougherty.
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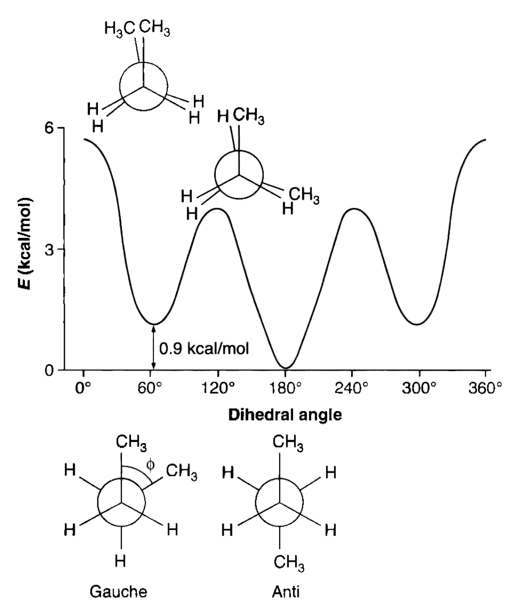

Figure 2: Second part of Figure 2.6 in "Modern Physical Organic Chemistry" by Anslyn & Dougherty.
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Table 2.14
Selected Cyclohexane *A* Values (in kcal/mol)*

Group	*A* value	Group	*A* value
D	0.006	CH_3	1.74
F	0.25–0.42	C_2H_5	1.79
Cl	0.53–0.64	$CH(CH_3)_2$	2.21
Br	0.48–0.67	$C(CH_3)_3$	4.7–4.9
I	0.47–0.61	CF_3	2.4–2.5
OH	0.60–1.04	C_6H_5	2.8
OCH_3	0.55–0.75	C_6H_{11}	2.2
OC_6H_5	0.65	CH_2Br	1.79
$OCOCH_3$	0.68–0.87	$Si(CH_3)_3$	2.5
$OSi(CH_3)_3$	0.74	$CH{=}CH_2$	1.5–1.7
NH_2	1.23–1.7	CHO	0.56–0.8
$N(CH_3)_2$	1.5–2.1	$COCH_3$	1.0–1.5
NO_2	1.1	CO_2^-	2.0
SH	1.21	CO_2H	1.4
SO_2CH_3	2.50	CO_2CH_3	1.2–1.3

*Eliel, E. L., Wilen, S. H., and Mander, L. N. (1994). *Stereochemisty of Organic Compounds*, John Wiley & Sons, New York.

Figure 3: Table 2.14 in "Modern Physical Organic Chemistry" by Anslyn & Dougherty.
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